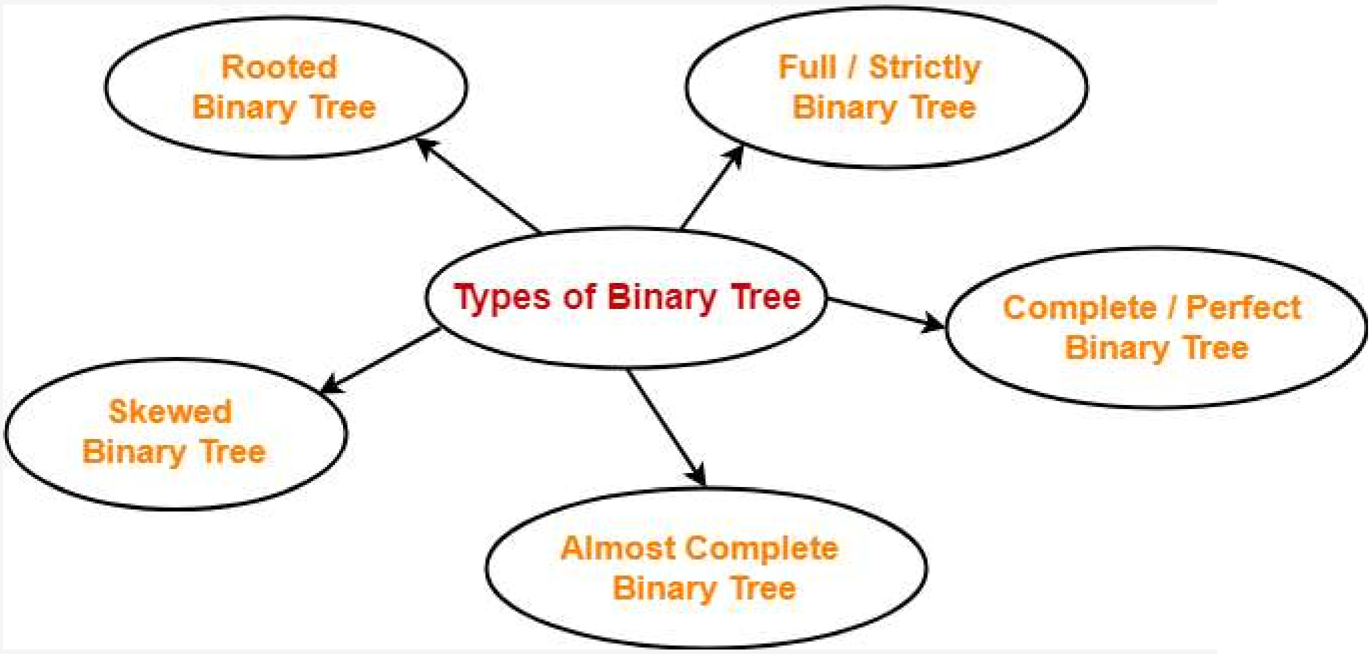
Binary Tree | Binary Tree Properties



Binary Tree Properties-

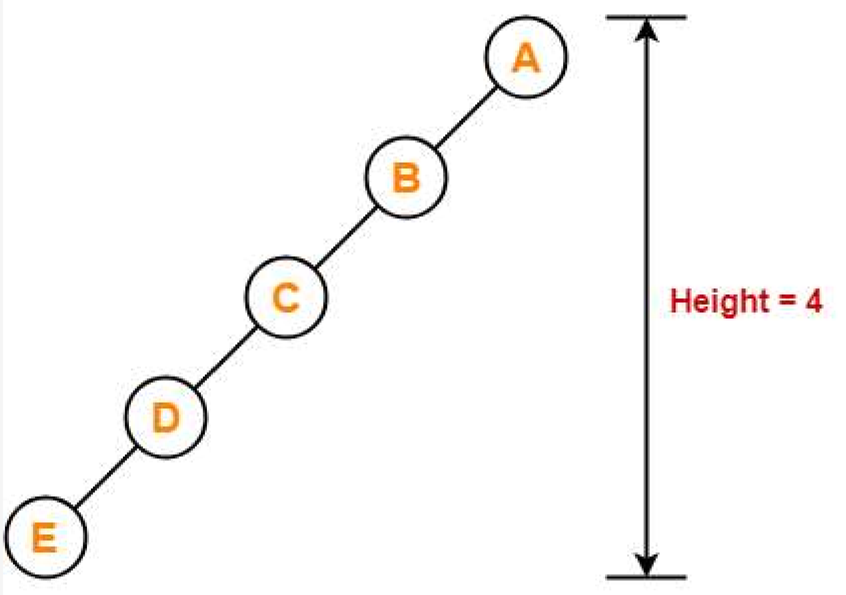
Important properties of binary trees are-

Property-01:

Minimum number of nodes in a binary tree of height H **= H + 1**

Example-

To construct a binary tree of height = 4, we need at least 4 + 1 = 5 nodes.



Property-02:

Maximum number of nodes in a binary tree of height H

= $2^{H+1} - 1$

Example-

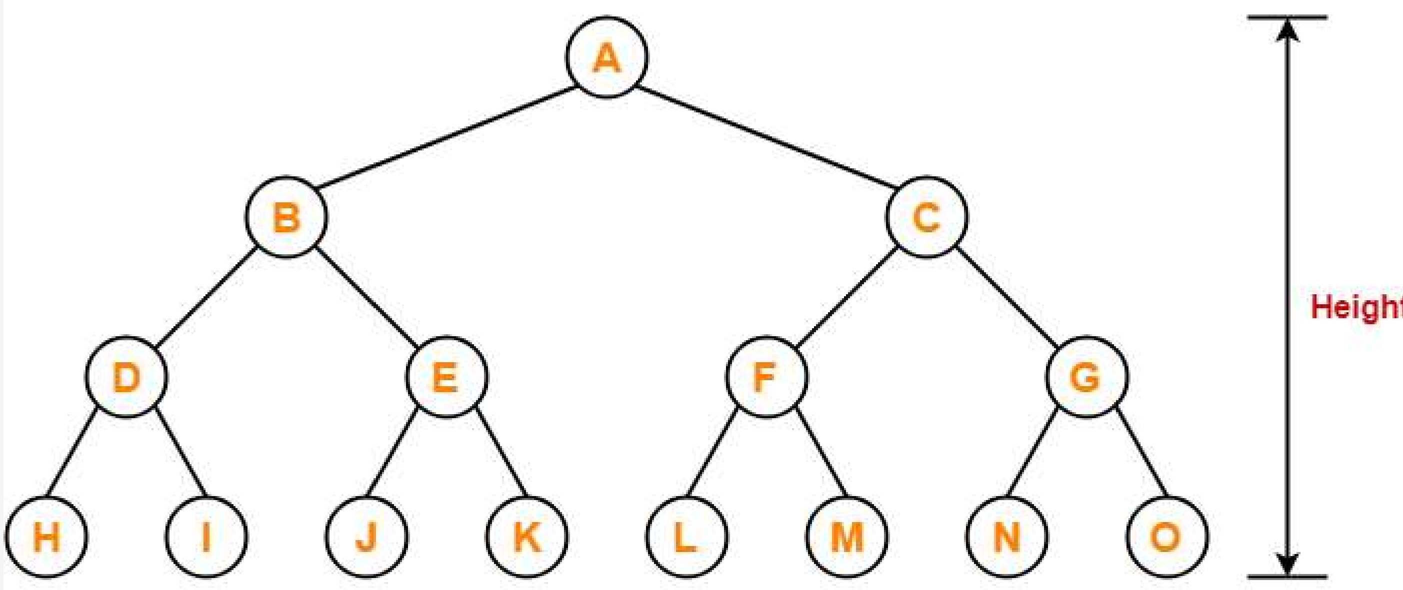
Maximum number of nodes in a binary tree of height 3

= $2^{3+1} - 1$

= 16 – 1

= 15 nodes

Thus, in a binary tree of height = 3, maximum number of nodes that can be inserted = 15.



We can not insert more number of nodes in this binary tree.

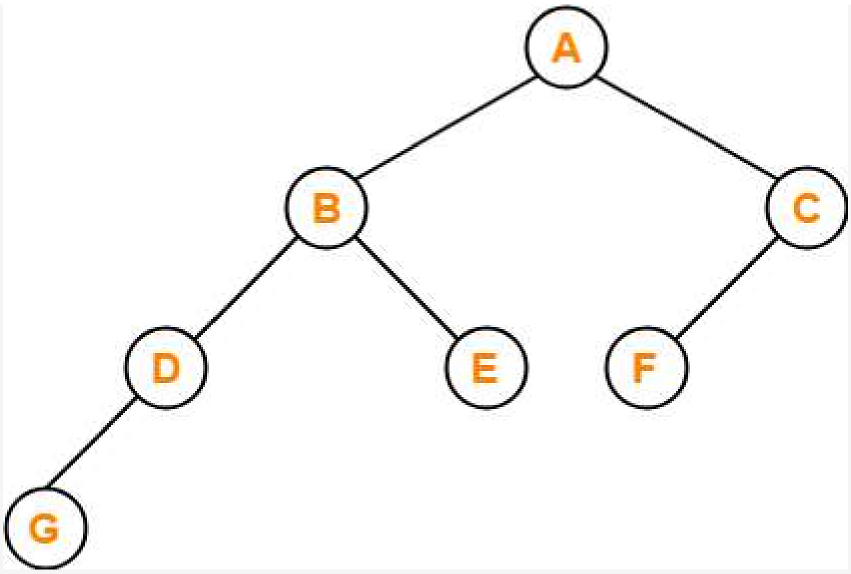
Property-03:

Total Number of leaf nodes in a Binary Tree

= Total Number of nodes with 2 children + 1

Example-

Consider the following binary tree-



Here,

- Number of leaf nodes = 3
- Number of nodes with 2 children = 2

Clearly, number of leaf nodes is one greater than number of nodes with 2 children.

This verifies the above relation.

NOTE

It is interesting to note that-

Number of leaf nodes in any binary tree depends only on the number of nodes with 2 children.

Property-04:

Maximum number of nodes at any level 'L' in a binary tree

= 2^L

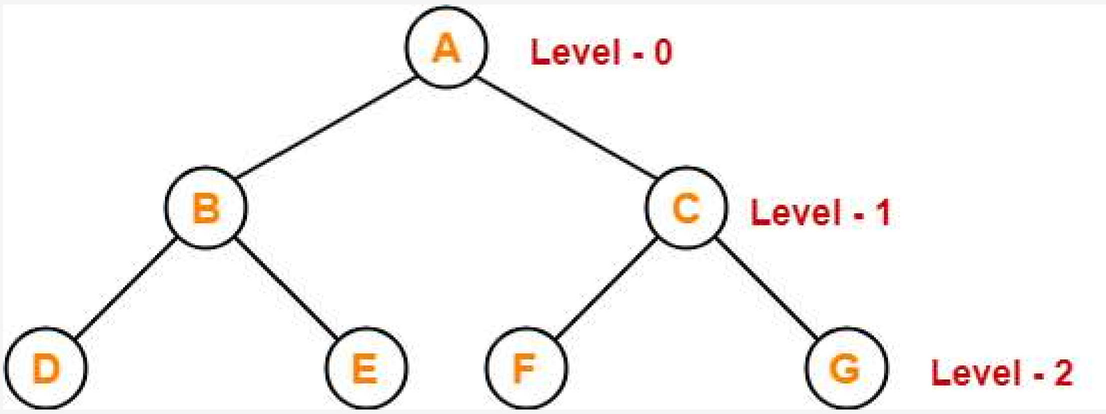
Example-

Maximum number of nodes at level-2 in a binary tree

$= 2^2$

$= 4$

Thus, in a binary tree, maximum number of nodes that can be present at level-2 = 4.



PRACTICE PROBLEMS BASED ON BINARY TREE PROPERTIES-

Problem-01:

A binary tree T has n leaf nodes. The number of nodes of degree-2 in T is ______?

A. $\log_2 n$
B. n-1
C. n
D. 2^n

Solution-

Using property-3, we have-

Number of degree-2 nodes

= Number of leaf nodes – 1

= n – 1

Thus, Option (B) is correct.

Problem-02:

In a binary tree, for every node the difference between the number of nodes in the left and right subtrees is at most 2. If the height of the tree is h > 0, then the minimum number of nodes in the tree is ______?

A. 2^{h-1}
B. $2^{h-1} + 1$
C. $2^h – 1$
D. 2^h

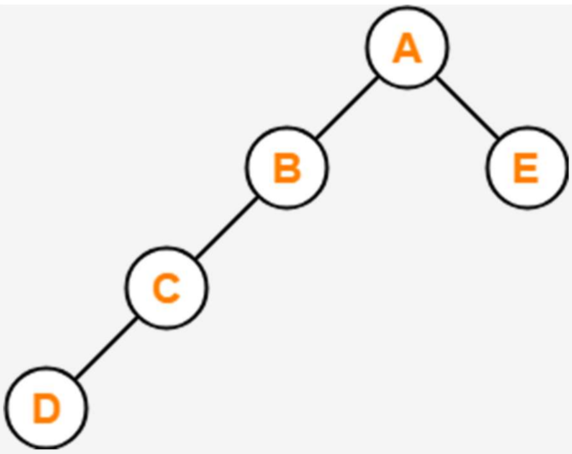
Solution-

Let us assume any random value of h. Let h = 3.

Then the given options reduce to-

A. 4
B. 5
C. 7
D. 8

Now, consider the following binary tree with height h = 3-



- This binary tree satisfies the question constraints.
- It is constructed using minimum number of nodes.

Thus, Option (B) is correct.

Problem-03:

In a binary tree, the number of internal nodes of degree-1 is 5 and the number of internal nodes of degree-2 is 10. The number of leaf nodes in the binary tree is ______?

A. 10
B. 11
C. 12
D. 15

Solution-

Using property-3, we have-

Number of leaf nodes in a binary tree

= Number of degree-2 nodes + 1

= 10 + 1

= 11

Thus, Option (B) is correct.

Problem-04:

The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is ______?

A. 2^h
B. $2^{h-1} – 1$
C. $2^{h+1} – 1$
D. 2^{h+1}

Solution-

Using property-2, Option (C) is correct.

Problem-05:

A binary tree T has 20 leaves. The number of nodes in T having 2 children is ______?

Solution-

Using property-3, correct answer is 19.